Emberz — speaker notes

SLIDE 1 · COVER

Emberz

Open: “Every one of us has a parent or grandparent who answers unknown calls. We built Emberz: a guardian that sits on their phone line, listens during the call and steps in before the scammer gets what they want.” Introduce yourselves: Numan built the live line, speech-to-text and the AI-voice detector; Matt built the guardian data, the fine-tune, the eval harness and the console.

SLIDE 2 · LOSSES

$7.75B

Older adults reported $7.75 billion in losses to the FBI last year, up 59%. The FTC thinks the true cost is $13 to $120 billion once unreported cases are counted. And the phone is where they lose the most: a scam that starts with a call costs a median $3,000, five times a scam that starts on social media. People over 80 lose more by phone than any other way.

SLIDE 3 · ATTACKER

AI made the scam call cheap, fluent and endless.

Here’s what changed. We built our own scam caller in an afternoon: an LLM script, an off-the-shelf neural voice, a cloud phone API. About 34 cents per two-minute call. On real phone calls it held a fluent, pressuring back-and-forth and asked for the verification code. That’s the attacker every family now faces, and the FBI already logged over 22,000 AI-related complaints last year.

SLIDE 4 · GAP

Filters judge the number before it rings. The damage happens during the conversation.

Carrier filters and caller-ID apps decide before you pick up, based on the number. A spoofed or unknown number gets through, and then nothing is watching while the caller works on Grandma for ten minutes. Emberz is built for that window: the conversation itself.

SLIDE 5 · PRODUCT

One guarded number. Four ways it steps in.

You give Grandma, your kid or yourself a guarded number. Every call passes through our line. When a caller starts working someone over — impersonating the bank, asking for the code, grooming, threats — Emberz does four things: whispers a coach line only the protected person hears; phones the caretaker, who can listen, talk or end the call with one key; follows rules the family writes in plain words; and flags AI voices.

SLIDE 6 · HOW

Every caller turn, scored live.

Under the hood: the call lands on our Telnyx line, both sides are transcribed live by NVIDIA Nemotron Speech Streaming running locally, and after every caller turn our fine-tuned guardian on Fireworks reads the conversation and the family's rules and returns a structured verdict. The same prompt is used in training, in the benchmark and on the live line, so what we measured is what runs. And a call is never left unguarded: if the guardian endpoint is down, it falls back to Nemotron Lightning on Fireworks serverless.

SLIDE 7 · LIVE

A real call, end to end

This isn't a mock-up. This is a replay of a real call from yesterday afternoon: our AI scam caller dialed the guarded number, which rang a real phone. Emberz raised an urgent alert when the caller impersonated the bank, whispered "Don't confirm anything, hang up and call the number on your card" only to Grandma, phoned the caretaker, and when the caller asked for the code, the caretaker pressed 9 and Emberz ended the call. [If time allows: play the 2-minute video, demo/video/renders/emberz_demo_final.mp4, or open demo/screens/replay.html.]

SLIDE 8 · SETUP

A fair test against a named frontier model.

Before the numbers, how we tested. 257 held-out calls, written by a different model family than our training data, so the test isn't the generator grading itself. The metric is the one that matters on a live call: did the guardian raise an alert before the caller asked for the code or the money? Same frozen prompt for every model, and every number comes with a 95% interval.

SLIDE 9 · HEADLINE

More calls handled right than Kimi K3.

Here's the headline. Count every call: a scam caught before the ask, or a harmless call left alone. Our fine-tuned guardian handles 90.3% of calls; Kimi K3, the frontier model, handles 83.7%. That gap is statistically significant: McNemar p of 0.003, and the bootstrap interval for the gain stays above zero even when we cluster by scenario.

SLIDE 10 · TABLE

Same catch rate, fewer false alarms.

The full picture. On catching scams, the fine-tune matches the best frontier models — the catch-rate gap with Kimi K3 is inside the error bars, and we say so. Where it wins clearly is everything else: fewer false alarms than every zero-shot model, less than half of Rune's and DeepSeek's, and it's the only model that reliably follows the family's own rules. Compare it with its own untuned base: the fine-tune cut false alarms from 53% to 18%.

SLIDE 11 · RULES

Only the fine-tune follows the family's rules.

This is the part frontier models just don't do. Every family is different: Grandpa is allowed to ask about the bank; the grandson is not allowed to ask for money. We test that with pairs: the same call, one rule changed, so the right answer flips. Our guardian gets 19 of 22 pairs right. Kimi K3 gets 4. A general model reads the call and ignores the family; ours reads the family's rules.

SLIDE 12 · ABLATE

What made it work.

Which choices mattered? Fine-tuning itself: the untuned base model alarms on more than half of harmless calls; one eight-minute, six-dollar LoRA run brought that to 16% and taught it the family's rules. More data (v2) raised the catch rate. Contrast pairs (v3) made it quieter but missed more. And if a family wants the quietest possible line, a cascade — Rune flags, our guardian confirms — gets false alarms down to 10% at a 90% catch rate.

SLIDE 13 · ROBUST

It holds up off-distribution, through real speech-to-text and against AI voices.

We also checked it outside our own data. On a public scam-conversation dataset it had never seen, it caught 98 of 100 scams with zero false alarms. When we voiced held-out calls over a phone-quality line and ran them through the same speech-to-text the live line uses, the catch rate didn't move. And our AI-voice detector — frozen wav2vec2 features plus a trained head — has a 1.27% equal error rate on speakers it never saw, and flagged 4 of our 5 live AI scam calls.

SLIDE 14 · COST

Cheap to run, fast enough to interrupt.

Who pays: adult children with an aging parent. What it costs us: about $3.64 per household per month for the phone line and the model, at scale — that's a model of the costs, with the assumptions in our business-value doc, not a measured bill. A frontier model on the same line would be about $9.37. It's fast enough: median under half a second per turn on a dedicated deployment. And the math for a family is simple: at ten dollars a month, stopping one typical phone-fraud loss every 25 years pays for it. [If asked: the cost model assumes one H100 at 50 concurrent calls; at low traffic, serverless frontier models are cheaper per minute.]

SLIDE 15 · COMPETE

Nobody else listens and calls the family.

Why not just use what exists? Google and Samsung warn the person on the phone — but the person being scammed is exactly the one who's been talked out of their judgment, and the family never hears. Carrier filters and Aura decide before the phone rings. Truecaller and Hiya check whether the voice is fake, not what the caller is asking for. Emberz is the only one that listens during the call and brings the family in, on the family's own rules.

SLIDE 16 · CONSENT

A model on the line has to earn its place.

The question we always get: isn't this creepy? We designed for that. Every call starts with a recorded notice, which all-party-consent states require. The protected adult opts in themselves; a caretaker can't enroll a competent adult. No voiceprints, and everything is deleted after seven days. And it stays quiet: a "watch" is only logged; the family is only called on a real alert, and the family's allowed contacts don't trigger it. Not legal advice — this is our design, documented in PLAN.md section 10.

SLIDE 17 · LIMITS

Honest limits.

We want to be straight about the limits. The deployed model loses some of its rule-following versus the checkpoint we evaluated; our hypothesis is the LoRA is only partly applied in serving, and we've raised it with Fireworks. All benchmark calls are generated, so real role-played calls are next. One-way recorded robocalls aren't what Emberz is for, and we're weaker than Kimi K3 there. And some samples are small; our clear wins are false alarms and the family's rules, not the raw catch rate.

SLIDE 18 · STACK

NVIDIA hears the call. Fireworks trains and serves the guardian.

Quick credit to the stack. NVIDIA Nemotron runs through the whole project: Speech Streaming hears every call, Nemotron 3 Ultra wrote our test set, we fine-tuned Nemotron 3.5 Super as a second guardian, and Nemotron Lightning is the fallback. Fireworks did the training — our first LoRA run took about eight minutes and six dollars — serves the guardian live, and hosted every baseline we compared against.

SLIDE 19 · CLOSE

Give Grandma a guarded number.

Close: “Scammers now have AI on their side for 34 cents a call. Emberz puts a guardian on the line, measured against a frontier model and working on real phones today. Give Grandma a guarded number. Thank you.” Then take questions. Likely questions: deployment drift (see limits slide), consent (every call starts with a recorded notice; the protected adult opts in themselves), cost at low traffic (serverless is cheaper until about 3–4 concurrent calls).